The documentation and process conversion measures necessary to comply with this document shall be completed by 22 May 2013.

INCH-POUND

MIL-PRF-19500/483D
w/AMENDMENT 1
22 February 2013
SUPERSEDING
MIL-PRF-19500/483D
2 April 2009

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, SILICON, HIGH-POWER,
THREE PHASE, FULL WAVE BRIDGE RECTIFIER,
TYPES M19500/483-01 THROUGH M19500/483-04, JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, 3-phase, full wave bridge rectifiers intended for use in applications at frequencies of 1 kHz or less. One level of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings. Unless otherwise specified $T_A$ = +25°C.

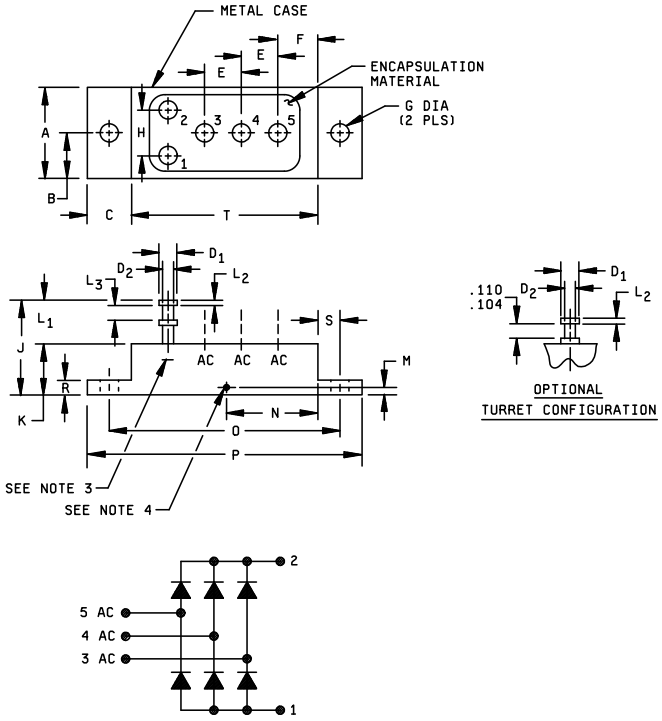
| Types | $V_{RWM}$ | Bridge (1) $I_{O1}$ at $T_C$ = +55°C | Bridge (2) $I_{O2}$ at $T_C$ = +100°C | $I_{F(surge)}$ $I_O = I_{O1}$ $T_C$ = +55°C $t_p$ = 8.3 ms | Barometric pressure (reduced) | $T_J$ | $T_{STG}$ |
|---|---|---|---|---|---|---|---|
| M19500/483- | V (pk) | A dc | A dc | A (pk) | mmHg | °C | °C |
| 01 | 200 | 25 | 18.5 | 150 | 8 | -65°C to +175°C | -65°C to +150°C |
| 02 | 400 | 25 | 18.5 | 150 | 8 | | |
| 03 | 600 | 25 | 18.5 | 150 | 33 | | |
| 04 | 800 | 25 | 18.5 | 150 | 33 | | |

(1) Derate from 25 A dc at $T_C$ = +55°C to 18.5 A dc at $T_C$ = +100°C (144 mA dc/°C).
(2) Derate from 18.5 A dc at $T_C$ +100°C to 0 A dc at $T_C$ = +150°C (370 mA dc/°C).

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .



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| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Ltr | Inches | | Millimeters | | Ltr | Inches | | Millimeters | |
| | Min | Max | Min | Max | | Min | Max | Min | Max |
| A | .710 | .770 | 18.03 | 19.56 | $L_1$ | .130 | .320 | 3.30 | 8.13 |
| B | .355 | .395 | 9.02 | 10.03 | $L_2$ | .015 | .030 | 0.38 | 0.76 |
| C | .355 | .395 | 9.02 | 10.03 | $L_3$ | .100 | .125 | 2.54 | 3.18 |
| $D_1$ | .141 | .151 | 3.58 | 3.84 | M | .040 | .060 | 1.02 | 1.52 |
| $D_2$ | .108 | .118 | 2.74 | 3.00 | N | .720 | .780 | 18.29 | 19.81 |
| E | .355 | .395 | 9.02 | 10.03 | O | 1.84 | 1.90 | 46.74 | 48.26 |
| F | .230 | .270 | 5.84 | 6.86 | P | 2.22 | 2.28 | 56.39 | 57.91 |
| G | .149 | .189 | 3.78 | 4.80 | R | .090 | .150 | 2.29 | 3.81 |
| H | .355 | .395 | 9.02 | 10.03 | S | .168 | .208 | 4.27 | 5.28 |
| J | | .820 | | 20.83 | T | 1.47 | 1.53 | 37.34 | 38.86 |
| K | .390 | .590 | 9.91 | 14.99 | | | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
* 3. Polarity shall be marked on the bridge body adjacent to terminals. Terminal numbers are for reference and are not required to be marked on the bridge.
4. Point at which $T_C$ is read shall be in metal part of case as shown on drawing.
5. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

* FIGURE 1. Physical dimensions.



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://assist.dla.mil/quicksearch or https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

N = number of diodes for each leg.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1.

3.4.1 Internal construction. The rectifier bridge shall consist of a metal and plastic encased assembly of single or parallel discrete diodes. Each discrete diode shall be a glass-to-metal, ceramic-to-metal, or fused metal oxide-to-metal hermetically sealed package. The silicon die in each discrete diode shall be metallurgically bonded directly to the terminal pins. The completed assembly of diodes and other internal structures shall be encapsulated in a plastic material which polymerizes to a rigid condition by virtue of a chemical cross-linking mechanism. The rectifier bridge shall be free of voids either visible or as evidenced by failure to pass the environment test specified. Only those discrete diodes which have met these requirements shall be used in the rectifier bridge. Discrete diodes shall be JANTX level. Parallel diodes may be utilized in construction of this device.

3.4.2 Terminal finish. Terminal finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of terminal finish is desired, it shall be specified in the acquisition document (see 6.2).



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3.4.3 Parallel diode construction. For bridge design that employs more than one diode for each leg and N equals the number of diodes for each leg, the following design and matching conditions shall be approved by the qualifying activity.

a. Individual diode room temperature $V_F$ matching shall be performed at a current value of ($I_{O1}$/N) or greater. The $V_F$ shall be within 20 mV on each diode.

b. The same diode design shall be used in each parallel leg.

c. The design shall ensure similar lead lengths and uniform heat paths, so that all diodes in each leg are at similar junction temperatures.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Polarity. Polarity shall be as marked on figure 1.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Screening (JANTX level). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I, subgroup 2 herein, and 4.3.1.1 herein shall not be acceptable.

4.3.1 Discrete diode screening. One hundred percent of the internal discrete diodes shall be subjected to the following:

| Screen (see table E-IV of MIL-PRF-19500) | Measurement |
|---|---|
| | JANTX level |
| 3c | Method 3101 of MIL-STD-750 (see 4.3.1.1) |
| 9 | Not applicable |
| 11 | $V_F$ at $I_F$/N and $I_{R2}$/N of subgroup 2 of table I herein |
| 12 | See 4.3.1.2 |
| 13 | $V_F$ at $I_F$/N and $I_{R2}$/N of subgroup 2 of table I herein. $\Delta V_{F1}$/n = ±0.05 V dc; $\Delta I_{R2}$ = 100 percent of initial value or ±250 nA dc, whichever is greater. Scope display evaluation (see 4.5.3) |



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4.3.1.1 Thermal impedance. The thermal impedance measurements shall be performed on the individual discrete diodes in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. The thermal impedance limit used in screen 3c of 4.3.1 and table I, subgroup 2 shall be set statistically by the supplier.

4.3.1.2 Power burn-in conditions. Power burn-in conditions shall be in accordance with method 1038 of MIL-STD-750, test condition B, $T_A$ = 55°C Maximum as defined in the general requirements of and mounting conditions in 4.5 of MIL-STD-750, $V_{RWM}$ = rated $V_{RWM}$ (see 1.3), $I_O$ = 3 A minimum, f = 50 - 60 Hz.

4.3.2 Assembly screening. One hundred percent of the completed assemblies shall be subjected to the following:

| Screen (see table E-IV of MIL-PRF-19500) | Measurement |
|---|---|
| | JANTX level |
| 3a | Method 1051 of MIL-STD-750, -55°C to +150°C, 10 cycles, no dwell required at +25°C. |
| 11 | Subgroup 2 of table I herein; dielectric withstanding voltage (see 4.5.1) |

4.4 Conformance inspection. Conformance inspection shall be performed on the finished rectifier assemblies in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIb (JANTX) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table E-VIb of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 1051 | Condition F, 25 cycles. |
| B2 | 1071 | Not applicable |
| B3 | 1027 | $I_O$ = 0 A dc; $T_A$ = +150°C; $V_{RWM}$ = 80 percent of rated $V_{RWM}$ (see 1.3); f = 60 Hz. Sampling plan n = 10 devices, c = 0. A separate sample may be pulled for each of the two B3 steps shown for simultaneous testing (TM1027 and operational thermal cycling). |
| B6 | | Not applicable |
| B7 | 3105 | See 4.5.2, n = 22, c = 0 |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.



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4.4.3.1 Group C inspection, table E-VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 1056 | Not applicable. |
| C2 | 2036 | Test condition A, 10 pounds, t = 30 seconds. |
| C2 | 2036 | Test condition D1, t = 10 seconds, 2 inch-pounds. |
| C2 | 1071 | Not applicable. |
| C5 | 1001 | Pressure = rated (see 1.3), $I_R$ = 1.0 μA dc maximum, $V_R$ = rated $V_R$ (see 1.3), terminal to case, sampling plan = 12 devices, c = 0. |
| C6 | 1026 | $I_0$ = 0 A dc; $T_A$ = +150°C; $V_{RWM}$ = 80 percent rated $V_{RWM}$ (see 1.3), f = 60 Hz. Sampling plan, n = 10, c = 0. |
| C7 | | Operational thermal cycling, (see 4.5.4); sampling plan, n = 10, c = 0. A separate sample may be pulled for C6 and C7 for simultaneous testing (method 1027 of MIL-STD-750 and operational thermal cycling). |
| C8 | 4066 | $V_{RWM}$ = rated $V_{RWM}$ (see 1.3), bridge $I_O$ = 25 A dc (entire bridge biased); $T_C$ = +55°C +10°C, -0°C; $I_{F(surge)}$ = 150 A (pk); $t_p$ = 8.3 ms; 10 surges for each leg at minimum 1 minute intervals; sampling plan = 22 devices, c = 0.<br><br>Alternate condition: Each leg may be biased separately to $I_O$ = 12.5 A dc, all other conditions the same as above. |

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Dielectric withstanding voltage. This test shall be performed with the metal case of the assembly connected to ground and all five terminals connected to the high potential side of a dc power supply or a scope display test set. The voltage applied between the terminals and the case shall be 2,800 volts, and shall be held at that level for 10 seconds. Any discontinuity or dynamic instability of the trace, or a breakdown current in excess of 10 μA dc, shall be cause for rejection.

4.5.2 Junction temperature test. The maximum $T_J$ of any diode in the bridge at the below specified conditions shall be ≤ +175°C. The test shall be performed in accordance with method 3105 of MIL-STD-750, except $V_{F4}$ shall be ≥ $V_{F2A}$. Test conditions shall be as follows:

a. $T_C$ = +55°C, $I_O$ = 25 A dc, $I_{ref}$ = 10 mA dc, or

b. $T_C$ = +100°C, $I_O$ = 18.5 A dc, $I_{ref}$ = 10 mA dc.



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4.5.3 Scope display evaluation. This test shall be conducted in accordance with method 4023 of MIL-STD-750. The following conditions shall apply:

a. Test condition B.

b. Horizonal sensitivity: 50 to 200V/division.

c. Vertical sensitivity: 5 to 20 µA/division.

d. Reverse current over the knee: 50 µA minimum.

4.5.4 Operational thermal cycling. This test shall be performed by connecting a three phase 60 Hz sine wave input of 10 to 20 V rms to the ac terminals of the bridge. The resistive load or input voltage shall be made variable in order to maintain a rated bridge output current of 25 A dc. Bridge case temperature shall be maintained at +55°C except during the cooling cycle when case temperature shall be permitted to drop 20°C maximum. Life-test duration shall be 2,000 cycles with a heating time of 8 minutes, +2, -0 minutes and a cooling time ≥ 2 minutes.

4.5.5 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | 4011 | | | | | |
| Forward voltage/leg | | | | | | |
| M19500/483-01 through -04 | | $I_F$ = 39 A dc (pk), (pulsed, see 4.5.5); $t_p$ = 8.3 ms maximum; duty cycle ≤ 2 percent | $V_{F1}$ | | 1.3 | V (pk) |
| Reverse current/leg | 4016 | DC method; $V_R$ = rated $V_{RWM}$ (see 1.3) | $I_{R1}$ | | 1.0 | µA dc |
| Reverse current (discrete diode) | 4016 | DC method; $V_R$ = rated $V_{RWM}$ (see 1.3) | $I_{R2}$ | | 1.0 | µA dc |
| Breakdown voltage/leg | 4021 | $I_R$ = 50 µA dc | $V_{BR}$ | | | |
| M19500/483-01 | | | | 220 | | V dc |
| M19500/483-02 | | | | 440 | | V dc |
| M19500/483-03 | | | | 660 | | V dc |
| M19500/483-04 | | | | 880 | | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C$ = +100°C | | | | |
| Reverse current/leg | 4016 | DC method; $V_R$ = rated $V_{RWM}$ (see 1.3) | $I_{R3}$ | | 200 | µA dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Reverse current/leg | 4016 | DC method; $V_R$ = rated $V_{RWM}$ (see 1.3) | $I_{R4}$ | | 2.0 | µA dc |
| Subgroup 4 | | | | | | |
| Reverse recovery time/leg; all devices | 4061 | Condition B, $I_F$ = 0.5 A, $I_R$ = 1.0 A, $I_{Rec}$ = 0.25 A | $t_{rr}$ | | 2.5 | µs |
| Dielectric withstanding voltage | | See 4.5.1 | | | | |
| Scope display | 4023 | See 4.5.3; n = 116, c = 0 | | | | |

1/ For sampling plan see MIL-PRF-19500.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Terminal finish (see 3.4.2).

d. Product assurance level and type designator.

e. Destructive physical analysis when requested.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil .

* 6.4 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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